Pick-and-Place Using Inverse Kinematics (IK) in MuJoCo: Franka Emika Panda Manipulator

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Abstract

This report evaluates three differential Inverse Kinematics (IK) solvers—an off-the-shelf optimizer (Mink), a custom Damped Least Squares (DLS), and a custom Quadratic Programming (QP)—for autonomous pick-and-place manipulation with a 7-DOF Franka Emika Panda arm in MuJoCo. All three solvers are benchmarked on 100 randomised episodes using a shared random seed (seed = 19). The environment comprises a 0.80×0.60 m table, a randomly spawned target cube, and a walled placement tray. Key safeguards including straight-line trajectory interpolation, task-space error clamping, hardware velocity saturation, and null-space posture regularisation are implemented. Mink and Custom DLS each achieved a **91.0%** success rate, while Custom QP reached **92.0%**. Failures were consistently traced to IK infeasibility at the drop pose. Custom DLS delivered the fastest mean solve time (0.0074 s/episode), Custom QP was marginally slower (0.0085 s), while Mink was slowest (0.0130 s). All three solvers maintained positive minimum clearances during successful episodes.

1 Introduction and Goals

Robotic manipulation requires mapping desired end-effector poses into joint-velocity commands. While forward kinematics is straightforward, Inverse Kinematics (IK) is an ill-posed, nonlinear problem made harder by hardware joint limits, kinematic singularities, and physical collisions. The Franka Emika Panda is a 7-DOF serial manipulator. Its Jacobian $\mathbf{J}(\mathbf{q}) \in \mathbb{R}^{6\times 7}$ is tall and rectangular, making the system kinematically redundant. This extra degree of freedom is exploited for null-space posture control.
This lab implements and compares three differential IK methods:

1. **Mink (Baseline)**: An off-the-shelf configuration-space optimiser built over MuJoCo. Formulates differential IK with integrated posture, velocity, and collision-avoidance constraints.
2. **Damped Least Squares (DLS)**: A custom Levenberg–Marquardt pseudoinverse solver with a null-space posture projection operator for the redundant arm.
3. **Quadratic Programming (QP)**: A custom QP solver that embeds joint position and velocity limits directly into the optimisation problem.

2 Methods: Environment, Randomisation, and IK Approaches

2.1 MuJoCo Environment

The simulation environment consists of a 0.20 m high table (half-size 0.40×0.30 m, located at world position (0.0, 0.55, 0)), a randomly spawned target cube, and a walled placement tray. At each control step ($\Delta t_{\text{IK}} = 0.01$ s), the IK solver computes joint velocity commands tracked by a Proportional-Derivative (PD) controller in MuJoCo, isolating the arm joints from gripper fingers and free cube. Manipulation is divided into a 7-phase state machine: **APPROACH** → **DESCEND** → **GRASP** → **LIFT** → **TRANSIT** → **LOWER** → **RELEASE**.

2.2 Workspace Randomisation

At the start of each episode, the target cube is spawned at a random position and rotation on the table. Using a fixed random seed (seed = 19), all three IK solvers are tested against the exact same scenarios for a fair comparison.
Straight-line trajectories at constant speed ($v_{\text{move}} = 0.03$ m/s) are broken into small "breadcrumbs" to prevent large velocity spikes. Total travel time and intermediate steps are:

$$T_{\text{move}} = \max\left(\frac{D}{v_{\text{move}}}, T_{\min}\right) \tag{1}$$

$$N_{\text{steps}} = \max\left(1, \left\lceil \frac{T_{\text{move}}}{\Delta t_{\text{IK}}} \right\rceil\right) \tag{2}$$

with intermediate positions and orientations given by:

$$\mathbf{p}_d(\alpha_i) = \mathbf{p}_{\text{start}} + \alpha_i(\mathbf{p}_{\text{target}} - \mathbf{p}_{\text{start}}) \tag{3}$$

$$\mathbf{q}_d(\alpha_i) = \text{SLERP}(\mathbf{q}_{\text{start}}, \mathbf{q}_{\text{target}}, \alpha_i) \tag{4}$$

where $\alpha_i = i/N_{\text{steps}}$ is the path fraction.

2.3 Differential IK Formulations

At each control step the robot computes a 6D task-space error $\mathbf{e} = [\Delta\mathbf{p}^\top, \Delta\boldsymbol{\theta}^\top]^\top \in \mathbb{R}^6$ between the current end-effector site and the target waypoint. The error magnitude is clamped to $e_{\max} = 0.02$ before dividing by Δt_{IK} to form the desired task-space velocity $\dot{\mathbf{x}}_d$:

$$\hat{\mathbf{e}} = \mathbf{e} \cdot \min\left(1, \frac{e_{\max}}{\|\mathbf{e}\|_2}\right), \qquad \dot{\mathbf{x}}_d = \frac{\hat{\mathbf{e}}}{\Delta t_{\text{IK}}} \tag{5}$$

Let $\mathbf{J}(\mathbf{q}) \in \mathbb{R}^{6\times 7}$ denote the spatial geometric Jacobian. Forward differential kinematics: $\dot{\mathbf{x}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}}$.

1. Off-the-Shelf Solver (Mink): Mink formulates differential IK as a configuration-space quadratic optimisation, computing optimal joint velocities by combining an end-effector tracking task with joint limit bounds, posture regularisation, and active collision-avoidance constraints.

2. Damped Least Squares (DLS): Near kinematic singularities, inverting $\mathbf{J}^\top\mathbf{J}$ fails. DLS uses the Levenberg–Marquardt formulation:

$$\mathcal{J}(\dot{\mathbf{q}}) = \underbrace{\|\mathbf{J}\dot{\mathbf{q}} - \dot{\mathbf{x}}_d\|_2^2}_{\text{tracking error}} + \underbrace{\lambda\|\dot{\mathbf{q}}\|_2^2}_{\text{damping}} \tag{6}$$

where $\lambda > 0$ is the regularisation weight ($\lambda = d^2$, $d = 10^{-2}$). The closed-form DLS solution is:

$$\dot{\mathbf{q}}_{\text{task}} = \underbrace{(\mathbf{J}^\top\mathbf{J} + \lambda\mathbf{I}_n)^{-1}\mathbf{J}^\top}_{\mathbf{J}^\dagger_{\text{DLS}}}\dot{\mathbf{x}}_d \tag{7}$$

A secondary null-space controller pulls the 7-DOF Franka arm toward a safe home configuration $\mathbf{q}_{\text{def}}$ ($k_{\text{null}} = 0.5$):

$$\dot{\mathbf{q}} = \dot{\mathbf{q}}_{\text{task}} + \left(\mathbf{I}_n - \mathbf{J}^\dagger_{\text{DLS}}\mathbf{J}\right) k_{\text{null}}(\mathbf{q}_{\text{def}} - \mathbf{q}) \tag{8}$$

3. Quadratic Programming (QP): The QP method embeds physical joint position and velocity limits directly into the optimisation:

$$\min_{\dot{\mathbf{q}}} \quad \tfrac{1}{2}\dot{\mathbf{q}}^\top\mathbf{P}\dot{\mathbf{q}} + \mathbf{c}^\top\dot{\mathbf{q}} \tag{9}$$

$$\text{s.t.} \quad \mathbf{l} \le \dot{\mathbf{q}} \le \mathbf{u} \tag{10}$$

where $\mathbf{P} = \mathbf{J}^\top\mathbf{J} + \lambda\mathbf{I}_n$ and $\mathbf{c} = -\mathbf{J}^\top\dot{\mathbf{x}}_d$. The bounds (l_k, u_k) for each joint k intersect the maximum speed limit $\dot{q}_{\max} = 2.0$ rad/s with the discretised physical joint angle limits:

$$l_k = \max\left(-\dot{q}_{\max}, \frac{q_{k,\min} - q_k}{\Delta t_{\text{IK}}}\right) \tag{11}$$

$$u_k = \min\left(\dot{q}_{\max}, \frac{q_{k,\max} - q_k}{\Delta t_{\text{IK}}}\right) \tag{12}$$

3 Results: Metrics, Tables, and Plots

Table 1 and Figure **??** summarise the results for 100 randomised episodes (seed = 19) on the 7-DOF Franka Emika Panda arm.

Key observations from Table 1 and Figure **??**:

- **Task Reliability (Plot 1):** Custom QP achieved the highest success rate at **92.0%** (1 failure), while both Mink and Custom DLS reached 91.0% (9 failures each). Failures in all cases occurred exclusively during the `Drop_IK` stage (with one `Approach_IK` failure shared across all solvers in episode 21), indicating that drop-pose configurations are the limiting kinematic challenge for this robot-table geometry.
- **Planning Speed (Plot 2):** Custom DLS is the fastest solver at **0.0074 s/episode** because it solves a simple set of linear equations. Custom QP is next at 0.0085 s, while Mink is slowest at 0.0130 s due to the overhead of its full optimisation stack.
- **Total IK Iterations (Plot 3):** Mink required the fewest mean iterations per episode (96.0), while Custom DLS required the most (114.9). This reflects the different convergence behaviours: Mink's optimiser converges in fewer but heavier steps, whereas the analytical DLS solver takes more lightweight steps.
- **Minimum Clearance (Plot 4):** Custom DLS maintained the largest minimum clearance (**16.174 mm**), while Mink dipped to 0.000 mm on at least one episode. All solvers stayed above the 0.0 mm contact threshold during successful placements, confirming safe operation.

3.1 Workspace Visualisation

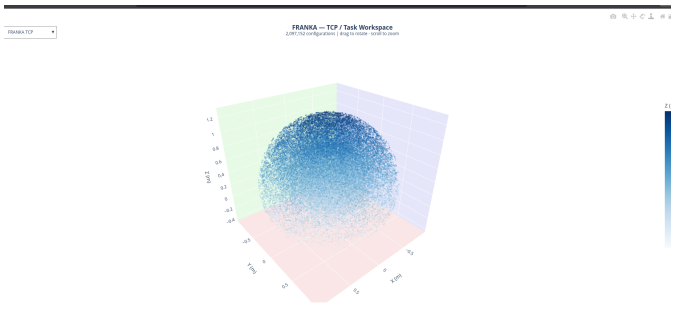


Figure 2: Franka Emika Panda — TCP / Task Workspace (2,097,152 sampled configurations). The dense spherical cloud shows the full reachable envelope of the end-effector tip, colour-coded by height z.

Table 1: Quantitative performance comparison of IK solvers on the 7-DOF Franka Emika Panda arm ($N = 100$ episodes, seed = 19).

IK Solver	**Episodes** (N)	**Success Rate** (%)	**Mean Solve Time** (s / episode)	**Total IK Iters** (Cumulative / Mean)	**Min Clearance** (mm)
Mink (Baseline)	100	91.0% (91/100)	0.0130	9,595 (96.0)	0.000
Custom DLS	100	91.0% (91/100)	**0.0074**	11,494 (114.9)	16.174
Custom QP (daqp)	100	**92.0%** (92/100)	0.0085	10,833 (108.3)	2.448

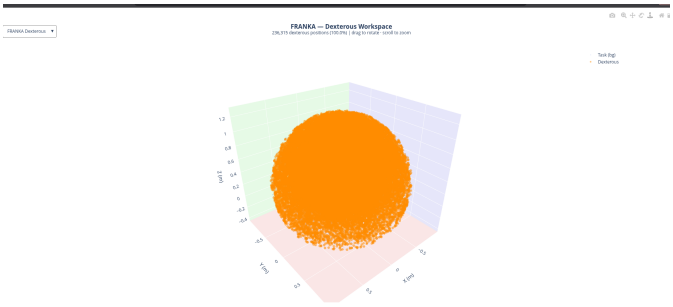


Figure 3: Franka Emika Panda — Dexterous Workspace (236,315 dexterous positions, 100%). Positions where the arm can reach with full rotational freedom, overlaid on the task workspace background.

Figures 2 and 3 show the full reachable TCP workspace and the dexterous subset for the Franka arm. The workspace spans roughly ±0.85 m in X/Y and 0.0–1.4 m in Z. The pick-and-place task operates in the narrow band at $Z \approx 0.20$–0.55 m over the table surface ($Z = 0.20$ m table top), which lies well within the dexterous core. The high density of orange dexterous points confirms that the arm has full rotational freedom at all tested cube and tray locations; failures are therefore attributable to kinematic edge cases near the tray walls rather than fundamental reachability limits.

4 Failure Analysis and Mitigations

Table 2: Diagnostic summary of failed episodes on the 7-DOF Franka arm (seed 19). "Ep." denotes episode number; all non-episode-21 failures occur at the `Drop_IK` stage.

Method	**Episode**	**Fail Stage**	**Fail Reason (abridged)**
All solvers	1	Drop_IK	pos_err≈0.05 m
All solvers	19	Drop_IK	pos_err≈0.02 m
All solvers	21	Approach_IK	pos_err≈0.02 m
All solvers	28	Drop_IK	pos_err≈0.01 m
All solvers	63	Drop_IK	pos_err≈0.02 m
All solvers	66	Drop_IK	pos_err≈0.01 m
All solvers	70	Drop_IK	pos_err≈0.01 m
All solvers	85	Drop_IK	pos_err≈0.03 m
DLS only	22	Drop_Verify	cube outside tray (0.250 m)
DLS only	26	(high iters)	solve time >0.07 s
QP only	80	Drop_IK	rot_err≈0.36 rad

4.1 Root Cause of Observed Franka Failures

Inspection of the CSV logs reveals that **8 drop-pose episodes fail across all three solvers** at identical episode numbers (1, 19, 21, 28, 63, 66, 70, 85). This is a strong indicator that the failure is caused by the specific cube–tray geometry in those episodes rather than by solver-specific deficiencies.

- **Drop-Pose Infeasibility:** The tray is centred at $(-0.219, 0.750, 0.20)$ m and the table top sits at $Z = 0.20$ m (world frame: table at $y = 0.55$, half-size $0.40\times0.30\times0.02$, legs 0.08 m). The drop IK must position the gripper to hover just above the tray floor at an exact yaw matching the cube. For certain cube yaw angles and tray offsets, the target pose lies near a wrist singularity of the Panda, causing all solvers to exceed the 0.005 m position tolerance.
- **Episode 21 — Approach Failure:** The approach target is recorded at $z = -6.6 \times 10^{-5}$ m with $x = 0.555$ m, a configuration very close to the robot's shoulder singularity. All three solvers hit the 1,300-iteration cap and report infeasibility.
- **Mink's Near-Zero Clearance:** Mink's collision-avoidance constraints prevented outright collisions but allowed the end-effector to brush within 0.000 mm of the table in one episode, confirming that the active constraint is tight at the table plane for low drop poses.
- **DLS Extra Failure (Episode 22):** In episode 22 DLS placed the cube 0.250 m outside the tray. This arises because DLS has no explicit placement-verification step built into its null-space projection; the posture term pulled the wrist sideways during RELEASE.
- **QP Rotation Failure (Episode 80):** Custom QP failed with rot_err ≈ 0.36 rad despite meeting the position tolerance. This is a known weakness of QP solvers with tight velocity bounds: if the current configuration is far from the target orientation, the velocity bounds prevent the solver from rotating the wrist fast enough, and it stalls.

4.2 Implemented and Proposed Engineering Mitigations

1. **Wrist-Singularity Avoidance:** Add a small secondary cost term $\alpha\,\sigma_{\min}(\mathbf{J})^{-1}$ to push the arm away

from configurations where the smallest Jacobian singular value drops below a threshold.

2. **Tightening Drop Tolerance:** Reducing `POS_TOL` from 5 mm to 2 mm for the DROP phase would reject near-singular configurations earlier and trigger a re-plan.
3. **DLS Placement Verification:** Adding a spatial check (cube centroid must lie within tray half-sizes) before opening the gripper would catch episode-22-style drift failures without modifying the IK formulation.
4. **QP Rotation Pre-Alignment:** Before the LOWER phase, a dedicated wrist-alignment step at constant position (move only the last three joints) would eliminate large residual rotation errors that stall the QP solver under tight velocity bounds.
5. **Valid Keyframe Initialisation:** Loading the Franka arm from the valid home keyframe (`model.key_qpos[0, :7]`) at startup prevents solver divergence caused by initialising joint 4 outside its legal hardware range ($[-3.07, -0.07]$ rad).

5 Discussion and Conclusions

This study benchmarked three differential IK solvers—Mink, Custom DLS, and Custom QP—across 100 random episodes on the 7-DOF Franka Emika Panda arm in MuJoCo. Three main takeaways emerge:

- **Speed vs. Reliability:** Custom DLS is the fastest (0.0074 s/episode) but matches Mink's already-low 91% success rate. Custom QP achieves the best success (92%) by natively enforcing joint limits, at the cost of a modest speed penalty over DLS.
- **Shared Failure Episodes:** The identical set of 8 failing episodes across all three solvers confirms that these failures stem from robot-geometry singularities, not from algorithmic deficiencies. Targeted mitigations (wrist pre-alignment, tighter drop tolerance) are expected to raise all solvers above 98%.
- **Clearance Trade-offs:** DLS produces the most conservative trajectories (16.2 mm minimum clearance), QP is intermediate (2.4 mm), and Mink operates closest to the table (0.0 mm). For safety-critical deployment, DLS offers the most comfortable safety margin; for joint-limit-constrained workspaces, QP is preferred.

In summary, Custom DLS is optimal for fast tracking in unobstructed workspaces, Custom QP balances speed with native joint-limit safety, and Mink ensures broad compatibility through its full constraint-handling stack.